A. Two-body decay with analytical integration over the parent azimuth

We consider an isotropic two-body decay

$$a \longrightarrow b + c \tag{1}$$

with branching ratio B. The Lorentz-invariant spectrum of the daughter particle b is related to the spectrum of the parent particle a by

$$E_p \frac{\mathrm{d}N_b}{\mathrm{d}^3 p} = \int \frac{\mathrm{d}^3 q}{(2\pi)^3\, 2E_q}\, D^a_{b|c}(p,q)\, E_q \frac{\mathrm{d}N_a}{\mathrm{d}^3 q}. \tag{2}$$

We use the mostly-positive metric convention

$$p \cdot q = -p^0 q^0 + \boldsymbol{p} \cdot \boldsymbol{q}. \tag{3}$$

For an isotropic two-body decay, the decay map is

$$D^a_{b|c}(p,q) = B\, \frac{4\pi^2 m_a}{p^a_{b|c}}\, \delta\left(q \cdot p + m_a E^a_{b|c}\right). \tag{4}$$

The energy and momentum of particle b in the rest frame of the parent are

$$E^a_{b|c} = \frac{m_a^2 + m_b^2 - m_c^2}{2m_a}, \tag{5}$$

$$p^a_{b|c} = \frac{\lambda^{1/2}(m_a^2, m_b^2, m_c^2)}{2m_a}, \tag{6}$$

where

$$\lambda(x,y,z) = x^2 + y^2 + z^2 - 2xy - 2xz - 2yz \tag{7}$$

is the Källén function.

1. Momentum parametrization

We parametrize the parent and daughter momenta as

$$q^\mu = (m_{aT} \cosh\eta_q,\, q_T \cos\phi_q,\, q_T \sin\phi_q,\, m_{aT} \sinh\eta_q)\,, \tag{8}$$

$$p^\mu = (m_{bT} \cosh\eta_p,\, p_T \cos\phi_p,\, p_T \sin\phi_p,\, m_{bT} \sinh\eta_p)\,, \tag{9}$$

where

$$m_{aT} = \sqrt{m_a^2 + q_T^2}, \qquad m_{bT} = \sqrt{m_b^2 + p_T^2}. \tag{10}$$

The variables η_q and η_p denote momentum rapidities.

The scalar product is

$$q \cdot p = -m_{aT} m_{bT} \cosh(\eta_q - \eta_p) + q_T p_T \cos(\phi_q - \phi_p). \tag{11}$$

Since

$$q_z = m_{aT} \sinh\eta_q, \qquad \mathrm{d}q_z = E_q\, \mathrm{d}\eta_q, \tag{12}$$

the invariant measure becomes

$$\frac{\mathrm{d}^3 q}{2E_q} = \frac{q_T}{2}\, \mathrm{d}q_T\, \mathrm{d}\eta_q\, \mathrm{d}\phi_q. \tag{13}$$

We introduce the invariant spectra

$$\mathcal{F}_a(q) \equiv E_q \frac{\mathrm{d}N_a}{\mathrm{d}^3 q}, \qquad \mathcal{F}_b(p) \equiv E_p \frac{\mathrm{d}N_b}{\mathrm{d}^3 p}. \tag{14}$$

Substitution into Eq. (2) gives

$$\begin{aligned}\mathcal{F}_b(p_T, \eta_p, \phi_p) = \frac{B m_a}{4\pi p^a_{b|c}} \int_0^\infty q_T \, \mathrm{d}q_T \int_{-\infty}^{+\infty} \mathrm{d}\eta_q \int_0^{2\pi} \mathrm{d}\phi_q \\ \times \delta\Big(- m_{aT} m_{bT} \cosh(\eta_q - \eta_p) + q_T p_T \cos(\phi_q - \phi_p) + m_a E^a_{b|c} \Big) \\ \times \mathcal{F}_a(q_T, \eta_q, \phi_q).\end{aligned} \tag{15}$$

2. *Analytical integration over the parent azimuth*

It is convenient to introduce the relative variables

$$\Delta\eta = \eta_q - \eta_p, \qquad \Delta\phi = \phi_q - \phi_p, \tag{16}$$

together with

$$M \equiv m_{aT} m_{bT}, \qquad Q \equiv q_T p_T, \qquad C \equiv m_a E^a_{b|c}. \tag{17}$$

The argument of the delta function can then be written as

$$f(\Delta\phi) = -M \cosh\Delta\eta + Q \cos\Delta\phi + C. \tag{18}$$

The condition $f(\Delta\phi) = 0$ gives

$$\cos\Delta\phi = \frac{M \cosh\Delta\eta - C}{Q}. \tag{19}$$

We therefore define

$$Z(q_T, \Delta\eta; p_T) \equiv \frac{M \cosh\Delta\eta - C}{Q}. \tag{20}$$

Real azimuthal roots exist only if

$$|Z| \le 1. \tag{21}$$

Defining

$$\alpha \equiv \arccos Z, \qquad 0 \le \alpha \le \pi, \tag{22}$$

the two azimuthal roots are

$$\phi_q^{(\pm)} = \phi_p \pm \alpha \qquad \mathrm{mod}\, 2\pi. \tag{23}$$

The derivative of the delta-function argument is

$$\frac{\partial f}{\partial \phi_q} = -Q \sin(\phi_q - \phi_p). \tag{24}$$

At either root,

$$\left| \frac{\partial f}{\partial \phi_q} \right|_{\phi_q = \phi_q^{(\pm)}} = Q\sqrt{1 - Z^2}. \tag{25}$$

Consequently,

$$\delta\big(f(\phi_q)\big) = \frac{\Theta(1-Z^2)}{Q\sqrt{1-Z^2}}\Big[\delta(\phi_q-\phi_p-\alpha) + \delta(\phi_q-\phi_p+\alpha)\Big]. \tag{26}$$

The azimuthal integration in Eq. (15) can therefore be performed analytically:

$$\mathcal{F}_b(p_T,\eta_p,\phi_p) = \frac{Bm_a}{4\pi p_{b|c}^a}\int_0^\infty q_T\,\mathrm{d}q_T\int_{-\infty}^{+\infty}\mathrm{d}\eta_q\,\frac{\Theta(1-Z^2)}{Q\sqrt{1-Z^2}} \times\Big[\mathcal{F}_a(q_T,\eta_q,\phi_p+\alpha)+\mathcal{F}_a(q_T,\eta_q,\phi_p-\alpha)\Big]. \tag{27}$$

Since $Q = q_T p_T$, this can equivalently be written as

$$\mathcal{F}_b(p_T,\eta_p,\phi_p) = \frac{Bm_a}{4\pi p_{b|c}^a p_T}\int_0^\infty \mathrm{d}q_T\int_{-\infty}^{+\infty}\mathrm{d}\eta_q\,\frac{\Theta(1-Z^2)}{\sqrt{1-Z^2}} \times\Big[\mathcal{F}_a(q_T,\eta_q,\phi_p+\alpha)+\mathcal{F}_a(q_T,\eta_q,\phi_p-\alpha)\Big]. \tag{28}$$

3. *Square-root representation*

For numerical purposes it is often preferable to avoid the ratio Z. Define

$$D(\Delta\eta) \equiv M\cosh\Delta\eta - C. \tag{29}$$

Then

$$Z = \frac{D}{Q}, \tag{30}$$

and

$$Q\sqrt{1-Z^2} = \sqrt{Q^2-D^2}. \tag{31}$$

Equation (27) becomes

$$\mathcal{F}_b(p_T,\eta_p,\phi_p) = \frac{Bm_a}{4\pi p_{b|c}^a}\int_0^\infty q_T\,\mathrm{d}q_T\int_{-\infty}^{+\infty}\mathrm{d}\eta_q \times\frac{\Theta\left(Q^2-D(\Delta\eta)^2\right)}{\sqrt{Q^2-D(\Delta\eta)^2}} \times\Big[\mathcal{F}_a(q_T,\eta_q,\phi_p+\alpha)+\mathcal{F}_a(q_T,\eta_q,\phi_p-\alpha)\Big], \tag{32}$$

where

$$\alpha = \arccos\left[\frac{M\cosh(\eta_q-\eta_p)-C}{Q}\right]. \tag{33}$$

4. *Fourier decomposition*

We can fourier decompose $\mathcal{F}_b(p_T,\eta_p,\phi_p) = \sum_n \mathcal{F}_{b,n}(p_T,\eta_p)e^{\mathrm{i}\phi_p n}$,we have

$$\mathcal{F}_{bn}(p_T,\eta_p) = \frac{Bm_a}{4\pi p_{b|c}^a p_T}\int_0^\infty \mathrm{d}q_T\int_{-\infty}^{+\infty}\mathrm{d}\eta_q\,\frac{\Theta(1-Z^2)}{\sqrt{1-Z^2}} \times\Big[2\cos(\alpha n)\Big]\mathcal{F}_{an}(q_T,\eta_q). \tag{34}$$

with It is convenient to introduce the relative variables

$$\Delta\eta = \eta_q - \eta_p, \qquad \Delta\phi = \phi_q - \phi_p, \tag{35}$$

together with

$$M \equiv m_{aT} m_{bT}, \qquad Q \equiv q_T p_T, \qquad C \equiv m_a E^a_{b|c}. \tag{36}$$

We therefore define

$$Z(q_T, \Delta\eta; p_T) \equiv \frac{M\cosh\Delta\eta - C}{Q}. \tag{37}$$

Real azimuthal roots exist only if

$$|Z| \leq 1. \tag{38}$$

Defining

$$\alpha \equiv \arccos Z, \qquad 0 \leq \alpha \leq \pi, \tag{39}$$

we have

$$\cos(n\alpha) = T_n(\cos(\alpha)) = T_n(Z) \tag{40}$$

$$\begin{aligned}\mathcal{F}_{bn}(p_T, \eta_p) =& \frac{B m_a}{4\pi p^a_{b|c} p_T} \int_0^\infty \mathrm{d}q_T \int_{-\infty}^{+\infty} \mathrm{d}\eta_q \, \frac{\Theta(1-Z^2)}{\sqrt{1-Z^2}} \\ & \times 2T_n(Z)\mathcal{F}_{an}(q_T, \eta_q).\end{aligned} \tag{41}$$

B. Three-body decay with analytical integration over the parent azimuth

We now consider the isotropic three-body decay

$$a \longrightarrow b + c + d. \tag{42}$$

The pair $c + d$ is treated as an effective particle $\widetilde{c}$ with invariant mass

$$\mu^2 = -(p_c + p_d)^2. \tag{43}$$

Its allowed mass range is

$$m_c + m_d \leq \mu \leq m_a - m_b. \tag{44}$$

For fixed μ, the energy and momentum of particle b in the rest frame of the parent are

$$E^a_b(\mu) = \frac{m_a^2 + m_b^2 - \mu^2}{2m_a}, \tag{45}$$

$$p^a_b(\mu) = \frac{\lambda^{1/2}(m_a^2, m_b^2, \mu^2)}{2m_a}. \tag{46}$$

The momentum of particle c in the rest frame of the effective particle $\widetilde{c}$ is

$$p^{\widetilde{c}}_c(\mu) = \frac{\lambda^{1/2}(\mu^2, m_c^2, m_d^2)}{2\mu}. \tag{47}$$

The three-body phase-space normalization is

$$\mathcal{N}_{a\to bcd} = \int_{m_c+m_d}^{m_a-m_b} \mathrm{d}\mu \, p^a_b(\mu) p^{\widetilde{c}}_c(\mu). \tag{48}$$

The effective three-body decay map is

$$D^a_{b|cd}(p,q) = \frac{4\pi^2 B m_a}{\mathcal{N}_{a\to bcd}} \int_{m_c+m_d}^{m_a-m_b} \mathrm{d}\mu\, \widetilde{p_c^c}(\mu)\, \delta\left(q\cdot p + m_a E_b^a(\mu)\right). \tag{49}$$

Using the same momentum parametrization as in the two-body case, the invariant daughter spectrum is

$$\begin{aligned}\mathcal{F}_b(p_T,\eta_p,\phi_p) = \frac{B m_a}{4\pi \mathcal{N}_{a\to bcd}} &\int_{m_c+m_d}^{m_a-m_b} \mathrm{d}\mu\, \widetilde{p_c^c}(\mu) \int_0^\infty q_T\, \mathrm{d}q_T \\ &\times \int_{-\infty}^{+\infty} \mathrm{d}\eta_q \int_0^{2\pi} \mathrm{d}\phi_q \\ &\times \delta\Big(-M\cosh\Delta\eta + Q\cos\Delta\phi + C_\mu\Big)\mathcal{F}_a(q_T,\eta_q,\phi_q),\end{aligned} \tag{50}$$

where

$$C_\mu \equiv m_a E_b^a(\mu). \tag{51}$$

For fixed μ, define

$$Z_\mu \equiv \frac{M\cosh\Delta\eta - C_\mu}{Q}. \tag{52}$$

The two azimuthal roots are

$$\phi_q^{(\pm)} = \phi_p \pm \alpha_\mu, \qquad \alpha_\mu = \arccos Z_\mu, \tag{53}$$

and they exist only when

$$|Z_\mu| \le 1. \tag{54}$$

The azimuthal delta-function decomposition is

$$\begin{aligned}\delta\Big(-M\cosh\Delta\eta + Q\cos\Delta\phi + C_\mu\Big) = &\frac{\Theta(1-Z_\mu^2)}{Q\sqrt{1-Z_\mu^2}} \\ &\times \Big[\delta(\phi_q-\phi_p-\alpha_\mu) + \delta(\phi_q-\phi_p+\alpha_\mu)\Big].\end{aligned} \tag{55}$$

Performing the ϕ_q integration gives

$$\begin{aligned}\mathcal{F}_b(p_T,\eta_p,\phi_p) = \frac{B m_a}{4\pi \mathcal{N}_{a\to bcd}} &\int_{m_c+m_d}^{m_a-m_b} \mathrm{d}\mu\, \widetilde{p_c^c}(\mu) \int_0^\infty q_T\, \mathrm{d}q_T \int_{-\infty}^{+\infty} \mathrm{d}\eta_q \\ &\times \frac{\Theta(1-Z_\mu^2)}{Q\sqrt{1-Z_\mu^2}} \\ &\times \Big[\mathcal{F}_a(q_T,\eta_q,\phi_p+\alpha_\mu) + \mathcal{F}_a(q_T,\eta_q,\phi_p-\alpha_\mu)\Big].\end{aligned} \tag{56}$$

Defining

$$D_\mu(\Delta\eta) \equiv M\cosh\Delta\eta - C_\mu, \tag{57}$$

one has

$$Q\sqrt{1-Z_\mu^2} = \sqrt{Q^2 - D_\mu(\Delta\eta)^2}. \tag{58}$$

The square-root representation is therefore

$$\begin{aligned}\mathcal{F}_b(p_T,\eta_p,\phi_p) = \frac{B m_a}{4\pi \mathcal{N}_{a\to bcd}} &\int_{m_c+m_d}^{m_a-m_b} \mathrm{d}\mu\, \widetilde{p_c^c}(\mu) \int_0^\infty q_T\, \mathrm{d}q_T \int_{-\infty}^{+\infty} \mathrm{d}\eta_q \\ &\times \frac{\Theta\left(Q^2 - D_\mu(\Delta\eta)^2\right)}{\sqrt{Q^2 - D_\mu(\Delta\eta)^2}} \\ &\times \Big[\mathcal{F}_a(q_T,\eta_q,\phi_p+\alpha_\mu) + \mathcal{F}_a(q_T,\eta_q,\phi_p-\alpha_\mu)\Big],\end{aligned} \tag{59}$$

where

$$\alpha_\mu = \arccos\left[\frac{M\cosh(\eta_q-\eta_p)-C_\mu}{Q}\right]. \tag{60}$$

1. Fourier analysis

We can solve in fourier space Performing the ϕ_q integration gives

$$\begin{aligned}\mathcal{F}_{bn}(p_T,\eta_p) = \frac{Bm_a}{4\pi\mathcal{N}_{a\to bcd}}\int_{m_c+m_d}^{m_a-m_b}\mathrm{d}\mu\, p_c^{\widetilde{c}}(\mu)\int_0^\infty q_T\,\mathrm{d}q_T\int_{-\infty}^{+\infty}\mathrm{d}\eta_q \\ \times\frac{\Theta(1-Z_\mu^2)}{Q\sqrt{1-Z_\mu^2}} \\ \times 2T_n(Z_\mu)\mathcal{F}_{an}(q_T,\eta_q).\end{aligned} \tag{61}$$

We use the Fourier convention

$$\mathcal{F}_a(q_T,\eta_q,\phi_q) = \sum_{n\in\mathbb{Z}}\mathcal{F}_{a,n}(q_T,\eta_q),e^{\mathrm{i}n\phi_q}, \qquad \mathcal{F}_{a,n} = \frac{1}{2\pi}\int_0^{2\pi}\mathrm{d}\phi_q, e^{-\mathrm{i}n\phi_q},\mathcal{F}_a, \tag{62}$$

and analogously for $\mathcal{F}_b$. Moreover,

$$Q = q_T p_T, \qquad Z_\mu = \frac{m_{aT}m_{bT}\cosh(\eta_q-\eta_p)-m_aE_b^a(\mu)}{Q}, \tag{63}$$

with

$$E_b^a(\mu) = \frac{m_a^2+m_b^2-\mu^2}{2m_a}. \tag{64}$$

Finally,

$$T_n(\cos\alpha) = \cos(n\alpha), \tag{65}$$

so that the two azimuthal roots give

$$e^{\mathrm{i}n\alpha}+e^{-\mathrm{i}n\alpha} = 2T_n(Z_\mu). \tag{66}$$

and

$$p_c^{\widetilde{c}}(\mu) = \frac{\sqrt{\lambda(\mu^2,m_c^2,m_d^2)}}{2\mu}, \qquad p_b^a(\mu) = \frac{\sqrt{\lambda(m_a^2,m_b^2,\mu^2)}}{2m_a}, \tag{67}$$

$$\mathcal{N}_{a\to bcd} = \int_{m_c+m_d}^{m_a-m_b}\mathrm{d}\mu, p_b^a(\mu), p_c^{\widetilde{c}}(\mu). \tag{68}$$